SCIE 1120 Chemistry and Life

遗传

DNA: The molecule of heredity

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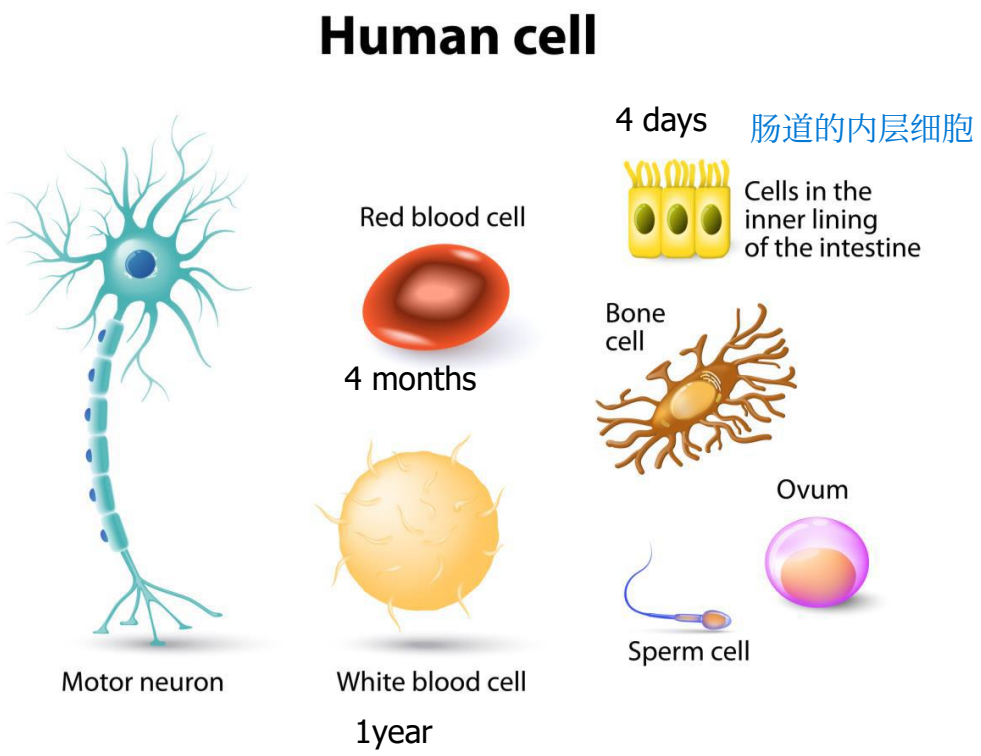
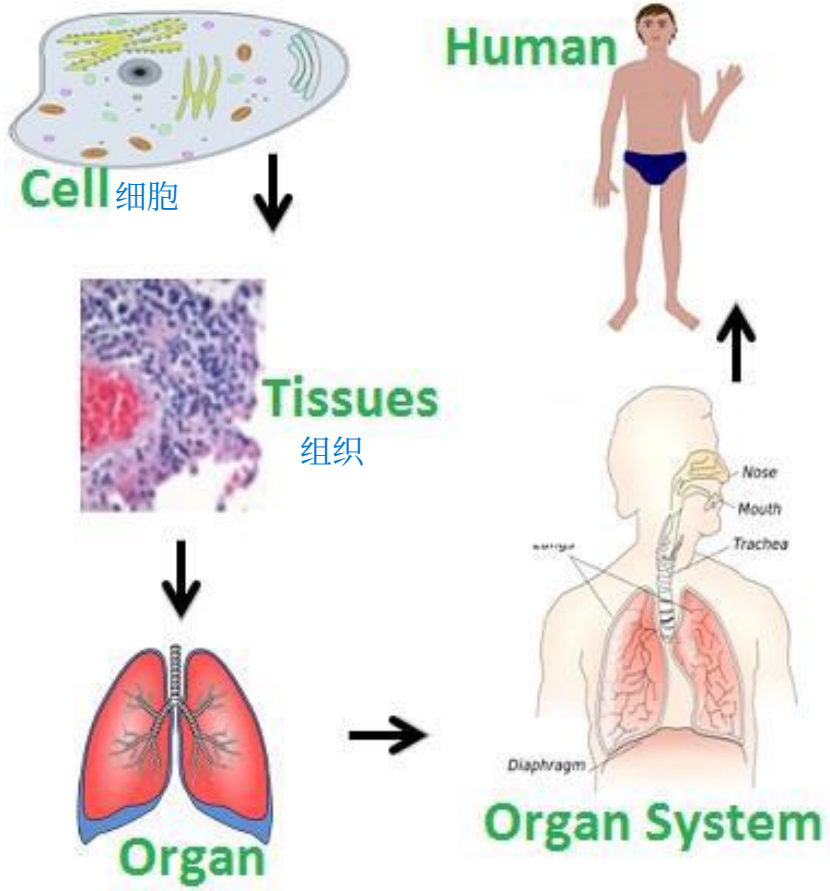
Apr.17, 2025, 3:00-4:20pm
LT-J

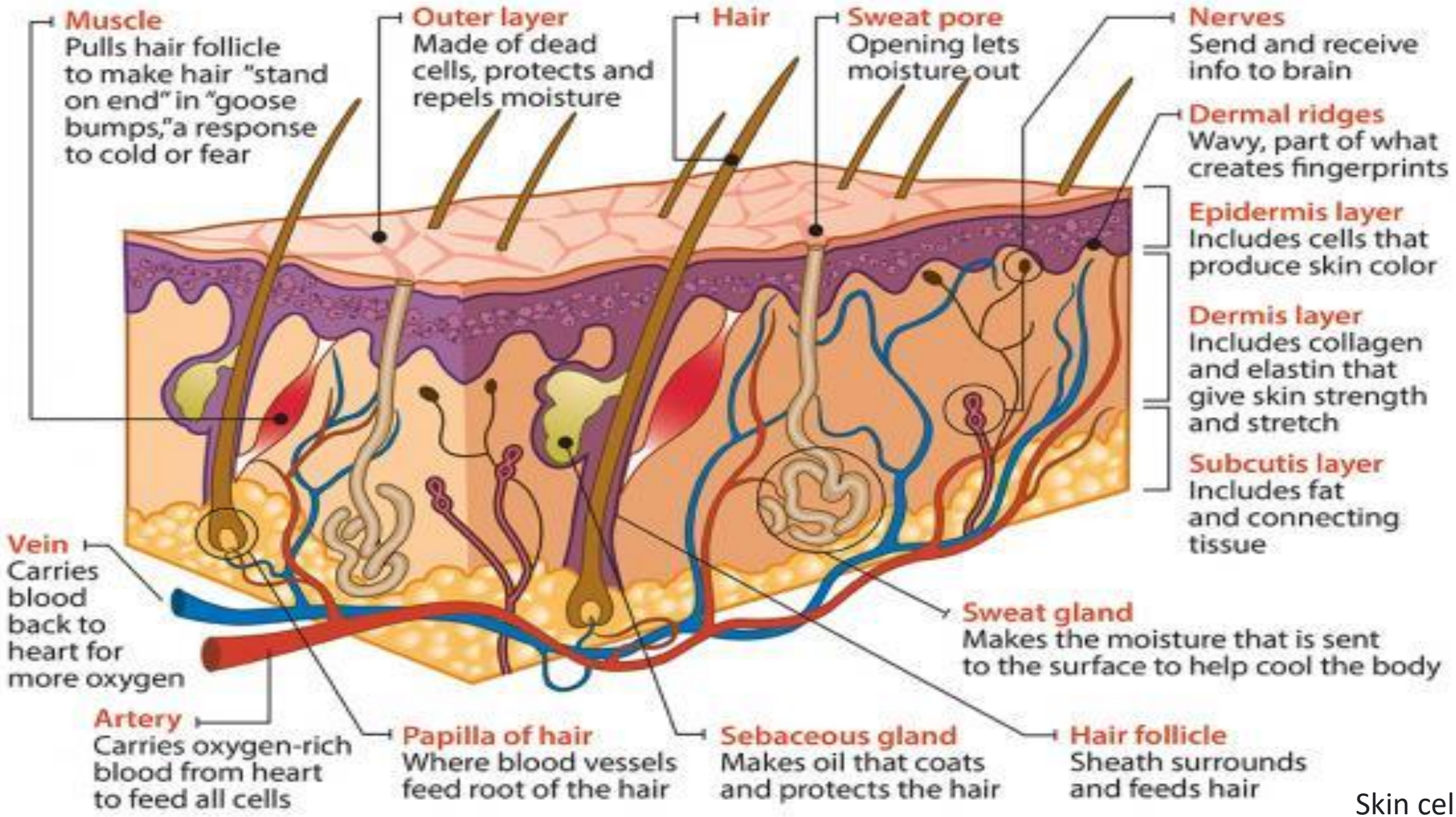
细胞是生物体的基本单位

Cells are Building Blocks of Living Organisms





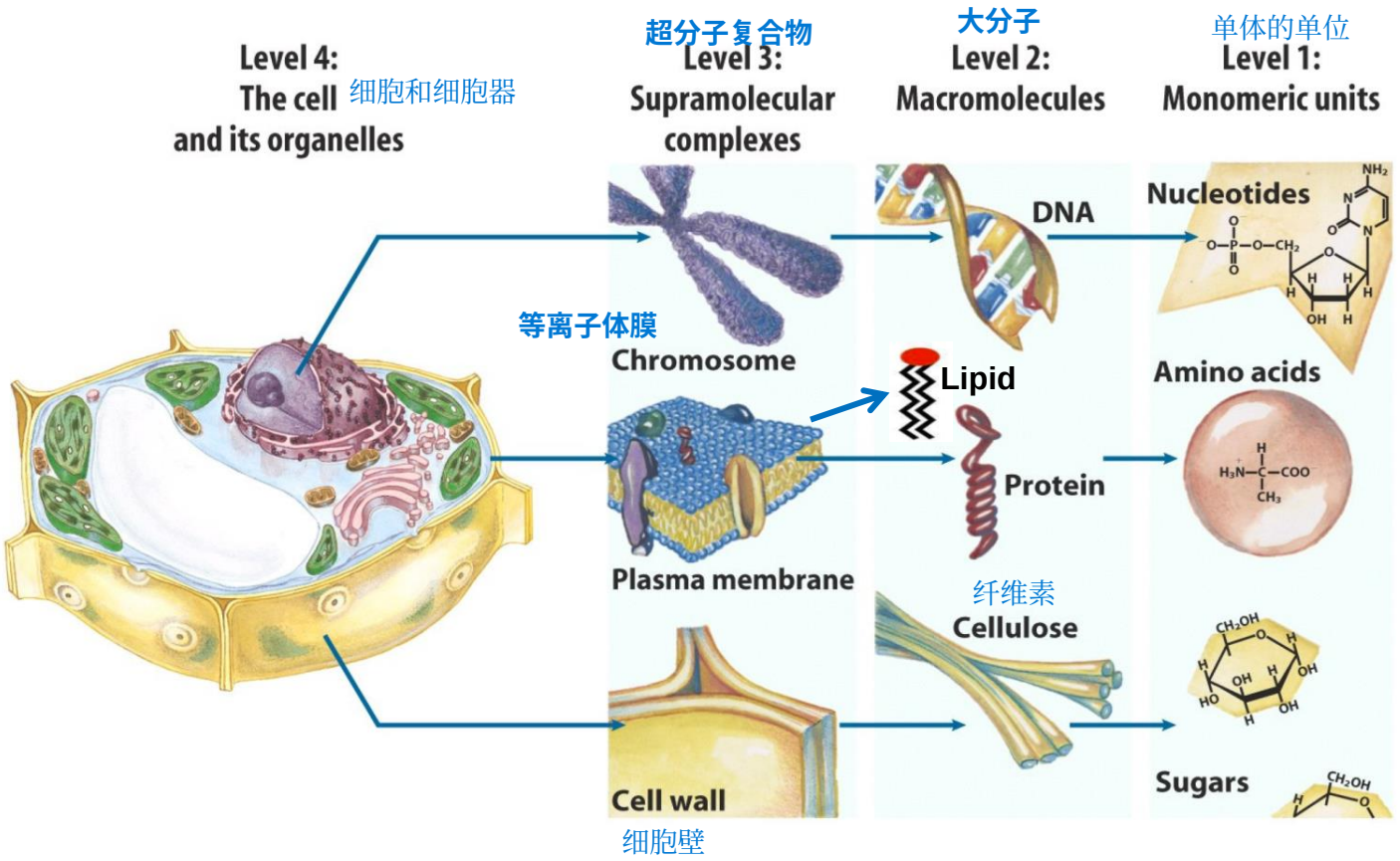
Lung Cell: 17 months



Skin cell:three weeks

The Molecules of Life

所有生物都由四类大分子生物物质构成：**碳水化合物、**脂质、蛋白质和核酸。大分子是由数千个共价键连接的原子组成的大**型分子。**分子的结构和功能是不可**分割的。**



Facts Of Evolution: The Molecules Of Life https://www.youtube.com/watch?v=nvJFI3ChOUU

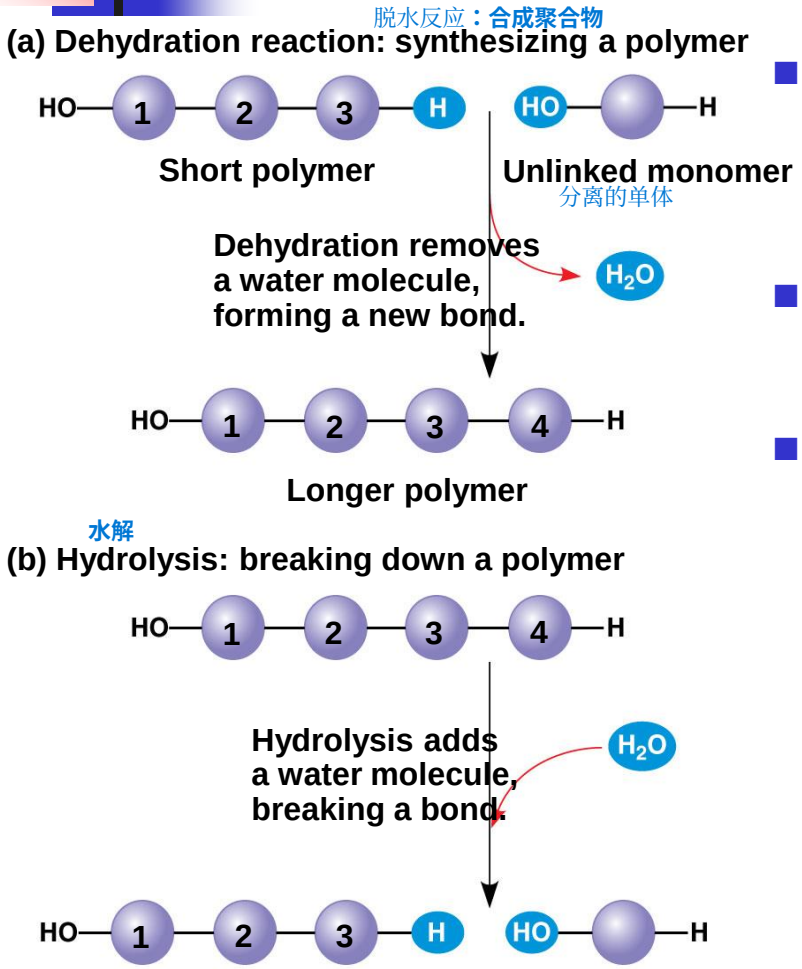
- All living things are made up of four classes of large biological molecules: carbohydrates, lipids, proteins, and nucleic acids
- Macromolecules are large molecules composed of thousands of covalently connected atoms
- Molecular structure and function are inseparable

大分子是由单体聚合而成的聚合物
Macromolecules are polymers, built from monomers

- A **polymer** is a long molecule consisting of many similar building blocks
- These small building-block molecules are called **monomers**
- Three of the four classes of life's organic molecules are polymers
  - Carbohydrates
  - Proteins
  - Nucleic acids

聚合物是由许多相似的结构单元组成的长链分子，这些小的结构单元分子被称为单体。
生命有机分子的四大类中有三类是聚合物
碳水化合物
蛋白质
核酸

Macromolecules are polymers built from monomers



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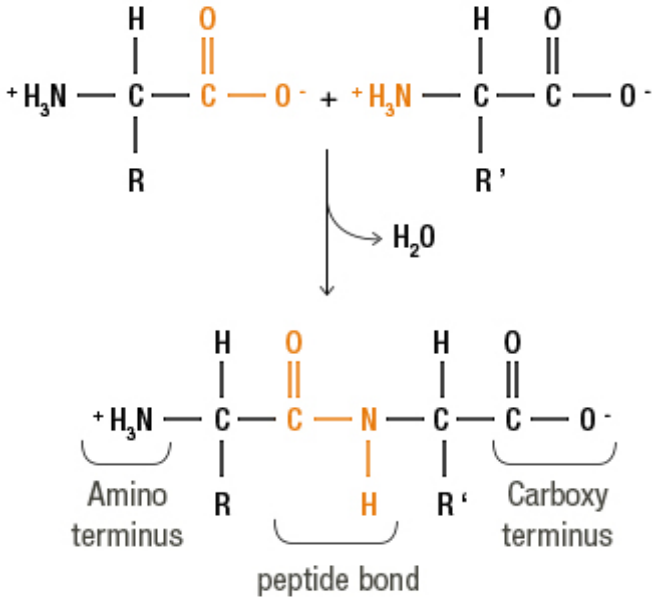


聚合物的多样性
The Diversity of Polymers

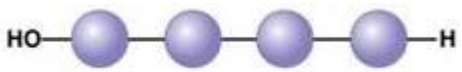
- Each cell has thousands of different macromolecules
- Macromolecules vary among cells of an organism, vary more within a species, and vary even more between species
- An immense variety of polymers can be built from a small set of monomers 仅用少量单体就能合成出种类繁多的聚合物

examples:

a protein of 100 amino acids can have 20^{100} possible linear arrangements.

a DNA of 50 nucleotides can have 4^{50} possible linear arrangements. (computer code is binary (2^{32} or 2^{64}))

由 100 个氨基酸组成的蛋白质可能有 20 到 100 种不同的线性排列方式。
一个由 50 个核苷酸组成的 DNA 可以有 4 的 50 次方种可能的线性排列

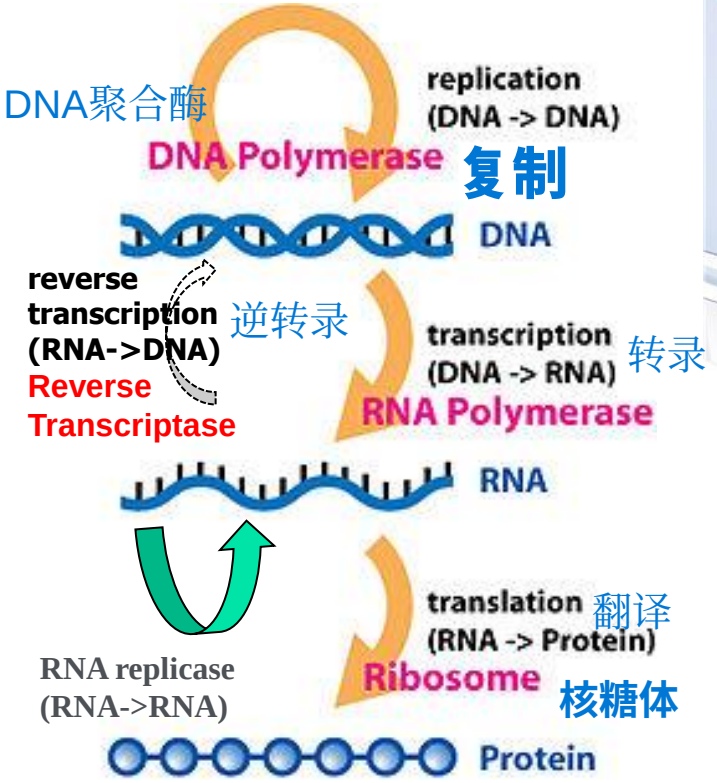


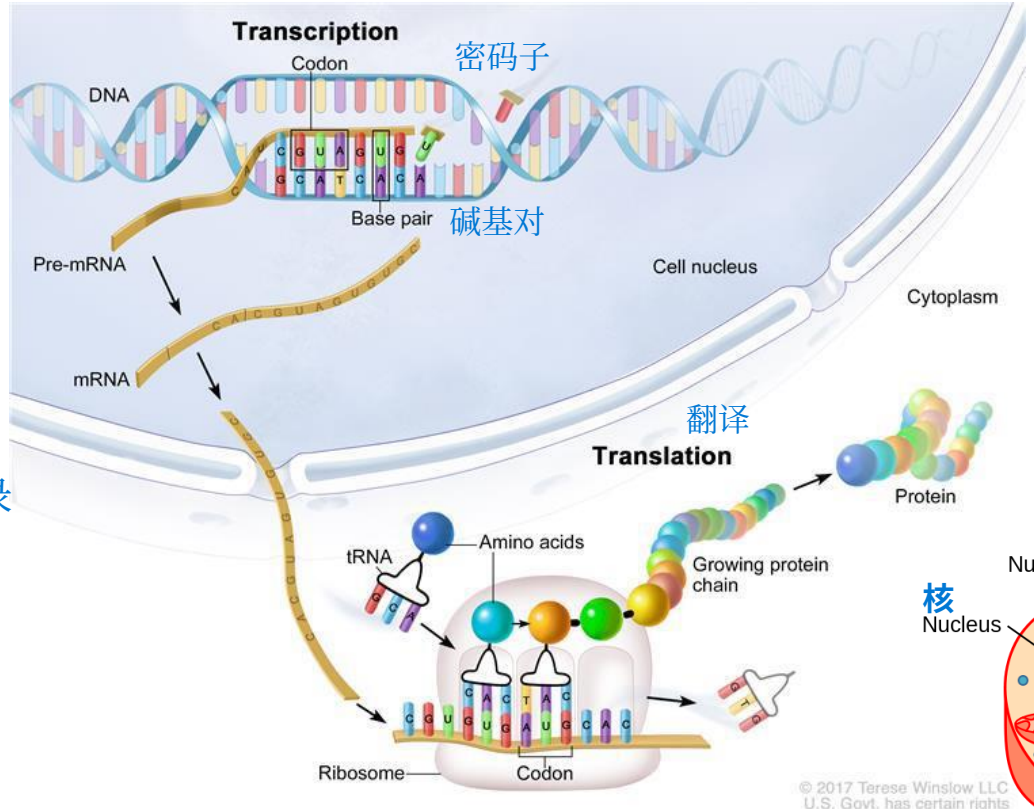
遗传分子

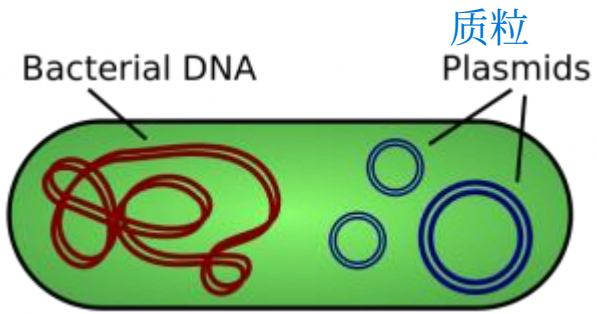
DNA: The molecule of heredity

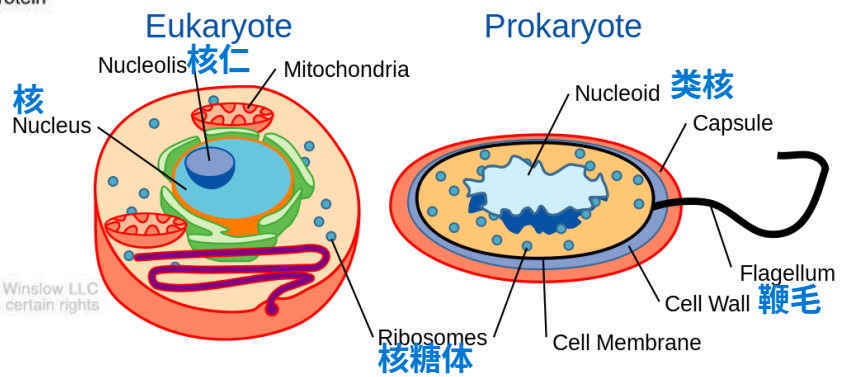
中心法则

Central dogma





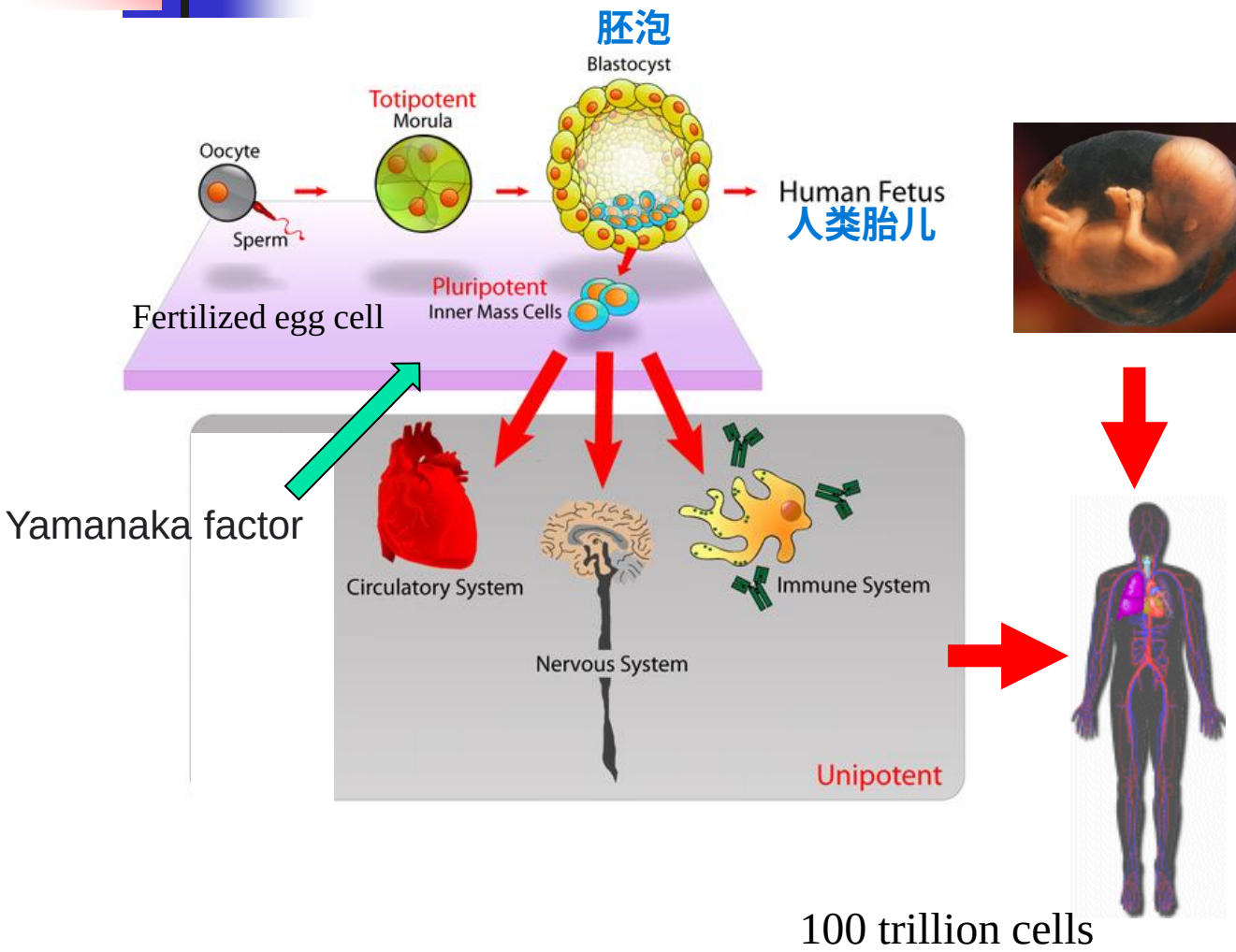




遗传信息

In eukaryotic cells, genetic instructions are housed in the nucleus

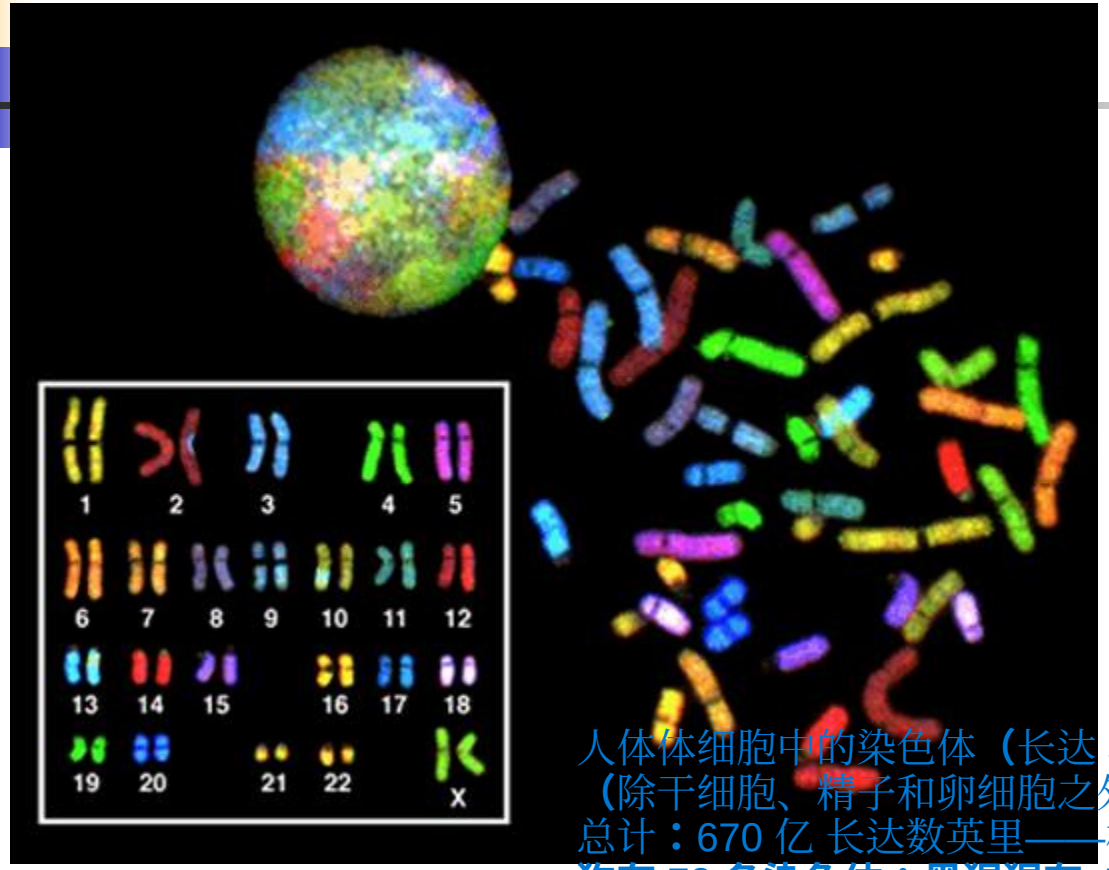
一个由单个受精卵发育而成的成年人，**体内有**100 万亿（100×10^12）个细胞。在每次细胞分裂过程中，遗传物质都必须被精确复**制。**
在人类的发育过程中，细胞分裂至少要进行一万亿次，任何一次失误都可能是致命的。
最终，复制缺陷开始显现。这最终可能与疾病和**衰老有关**



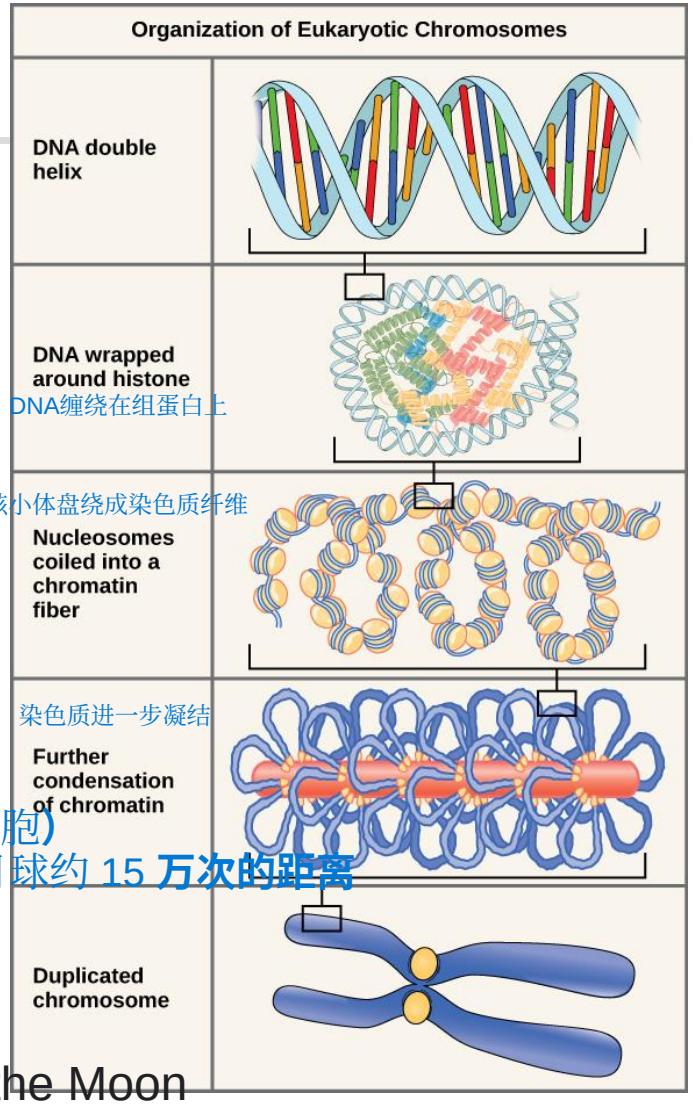
- A full-grown person consisting 100 trillion ($100x10^{12}$) cells from a single fertilized egg cell. During each cell division the genetic material has to be copied exactly.
- During the development of a human, there are at least trillion numbers of cell divisions, any mistake could be fatal.
- Eventually, replication defect kicks in. This could ultimately relate to diseases and aging
- During animal development, many cells become more specialized, the cellular differentiation which requires many different proteins. But their genome remain same.

在动物发育过程中，许多细胞变得更加特化，这种细胞分化需要多种不同的蛋白质。但它们的基因组保持不变。

DNA organization in eukaryotic cells



人体体细胞中的染色体（长达 3 米）
（除干细胞、精子和卵细胞之外的生物体细胞）
总计：670 亿 长达数英里——相当于往返月球约 15 **万次的距离**
狗有 78 **条染色体；黑猩猩有** 48 **条染色体。**

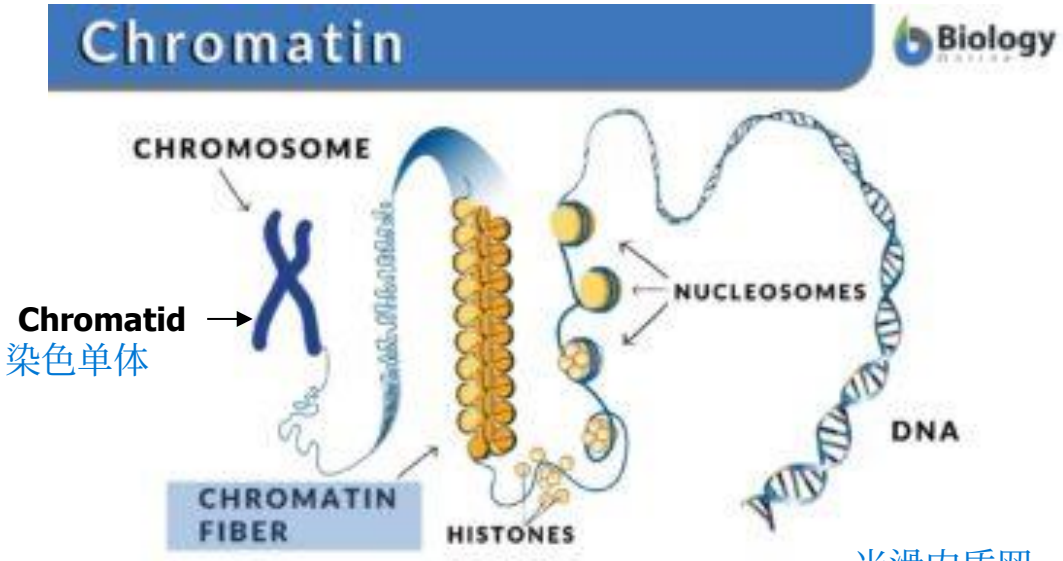


Chromosomes in a human somatic cell (3 meters long)
(cells of a living organism other than the stem, sperm and egg cells)
Total:67 billion miles long—the same as about 150,000 round trips to the Moon

Dogs have 78 chromosomes; chimpanzee has 48 chromosomes

DNA organization

染色质



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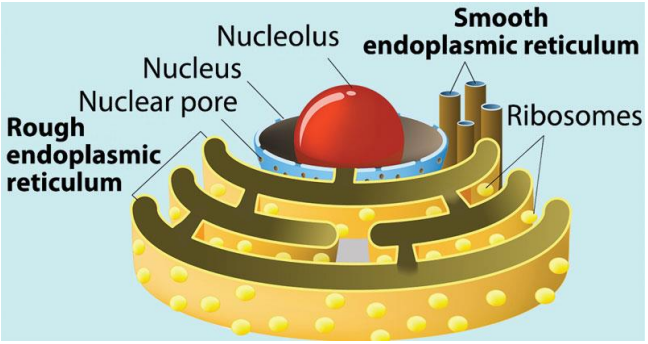

- In the nucleus, DNA is organized into discrete units called **chromosomes.**
- Each chromosome is composed of a single DNA molecule associated with proteins.
- The DNA and proteins of chromosomes are together called **chromatin.**
- Chromatin condenses to form discrete. **chromosomes** as a cell prepares to divide.
- The **nucleolus** is located within the nucleus and is the site of ribosomal RNA (rRNA) synthesis.

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Ribosomes: Protein Factories

- **Ribosomes** are particles made of ribosomal RNA and protein.
- Ribosomes carry out protein synthesis in two locations:
  - In the cytosol (free ribosomes) 胞浆
  - On the outside of the endoplasmic reticulum or the nuclear envelope (bound ribosomes)
    核膜上的内质网外侧

DNA could store all of the world's data in one room
1.8 bits of data per nucleotide due to errors

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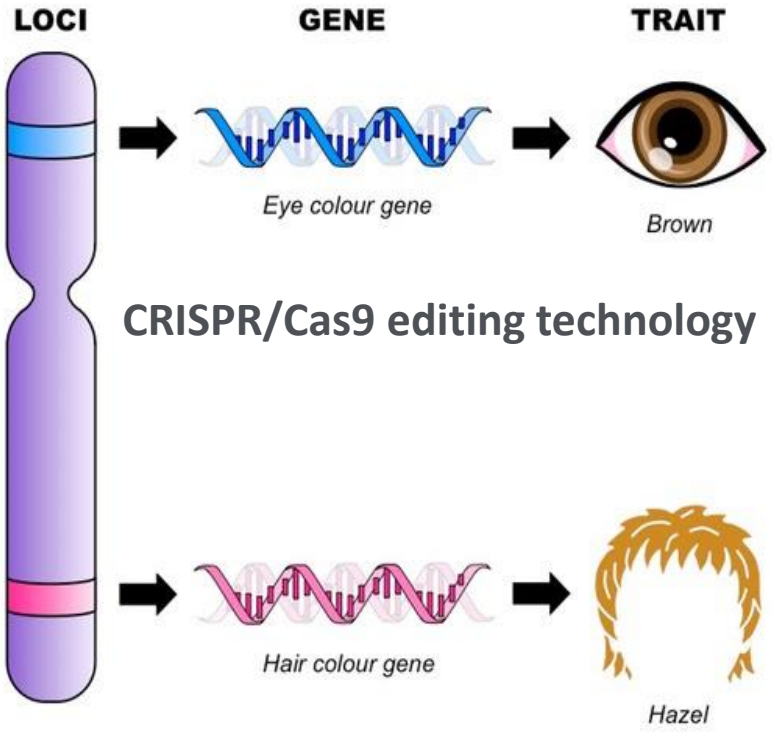
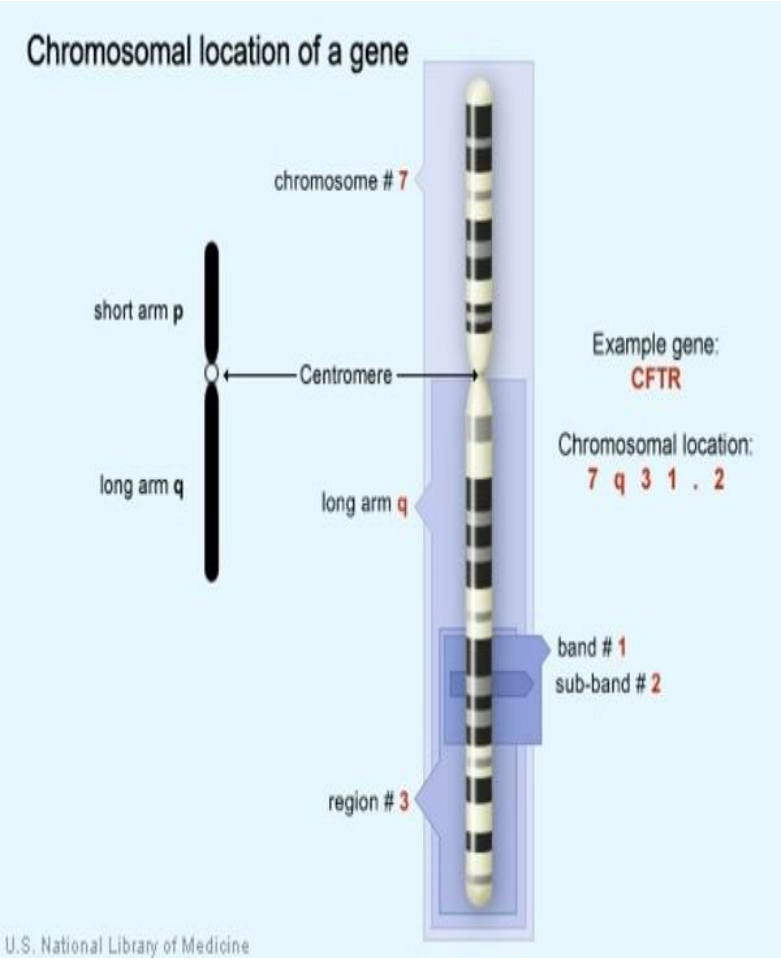
Basic definitions

基因是 DNA **的一段序列，**它编码特定的性状**（性状也可能受多个基因的影响）**（人类约有 2 **万个基因）**
基因在特定染色体上的位置称为基因座（复数形式为基因座）
•等位基因是基因的一种变体形式（比如眼睛的颜色）。人类被称为二倍体**（倍性=染色体数）生物，**因为在每个基因位点上都有两个等位基因，**分别从父母双方各**继承一个。
同源染色体由长度、着丝粒位置和染色模式大致相同的一对染色体组成，**其基因**在相同的位置上。一条同源染色体来自生物体的母亲**；另一条来自生物体的父**亲。



•A **gene** is a sequence of DNA that encodes for a specific trait (traits may also be influenced by multiple genes) (20K genes in human)
•The position of a gene on a particular chromosome is called the **locus** (plural = loci)
•An **allele** is a variant form of a gene (such as eye color). Humans are called diploid (ploidy=#chromosomes) organisms because they have two alleles at each genetic locus, with one allele inherited from each parent.
等位基因allele **同源染色体**homo

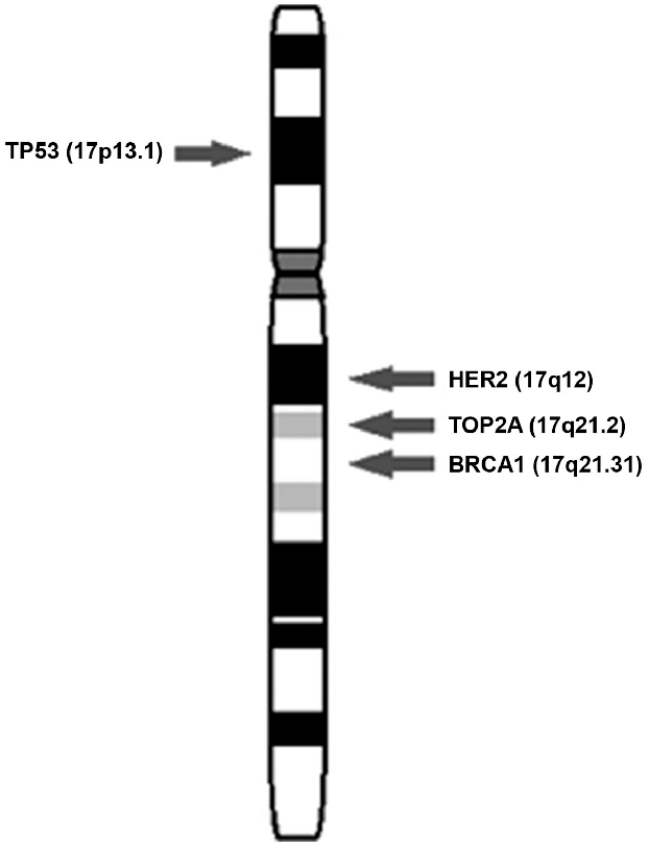
Homologous chromosomes are **made up of chromosome pairs of approximately the same length, centromere position, and staining pattern, for genes with the same corresponding loci**. One homologous chromosome is inherited from the organism's mother; the other is inherited from the organism's father.



HER2（人类表皮生长因子受体 2）**是一种基因，**可能在乳腺癌的发展**（增殖）**中起作用。病理报告应包含有关HER2 状态的信息。在大约 10% **至** 20% **的乳腺癌病例中，**HER2 **基因不能正常工作，**并且会过度复制自身。所有这些 **多余的** HER2 基因会指示乳腺细胞生成过多的 HER2 **受体**（HER2 蛋白过度表达）

基因包含细胞保持健康和正常运作所需的各种蛋白质的配方。某些基因及其所制造的蛋白质会影响乳腺癌的表现以及对特定治疗的反应。可以对组织样本中的癌细胞进行检测，以查看哪些基因正常，哪些异常。这些基因所制造的蛋白质也可以进行检测

HER2 gene in chromosome 17



- *HER2 (human epidermal growth factor receptor 2)* is a gene that can play a role in the development of breast cancer (proliferation). Pathology report should include information about **HER2 status**. In about 10% to 20% of breast cancers, the *HER2* gene doesn't work correctly and makes too many copies of itself. All these extra *HER2* genes tell breast cells to make too many HER2 receptors (***HER2 protein overexpression***)
- Genes contain the recipes for the various proteins for a cell needs to stay healthy and function normally. Some genes and the proteins they make can influence how a breast cancer behaves and how it might respond to a specific treatment. Cancer cells from a tissue sample can be tested to see which genes are normal and abnormal. The proteins they make can also be tested.

Angelina Jolie

Life's Operating Instructions

- In 1953, James Watson and Francis Crick introduced an elegant double-helical model for the structure of deoxyribonucleic acid, or DNA
- DNA, the substance of inheritance, is the most celebrated molecule of our time
- physiological hereditary information is encoded in DNA and reproduced in all cells of the body
- This DNA program directs the development of biochemical, anatomical, physiological, and (to some extent) behavioral characters (characters, including selfishness)

Structure of Double Helix DNA by X-ray Crystallography

通过 X 射线晶体学解析的双螺旋 DNA 结构

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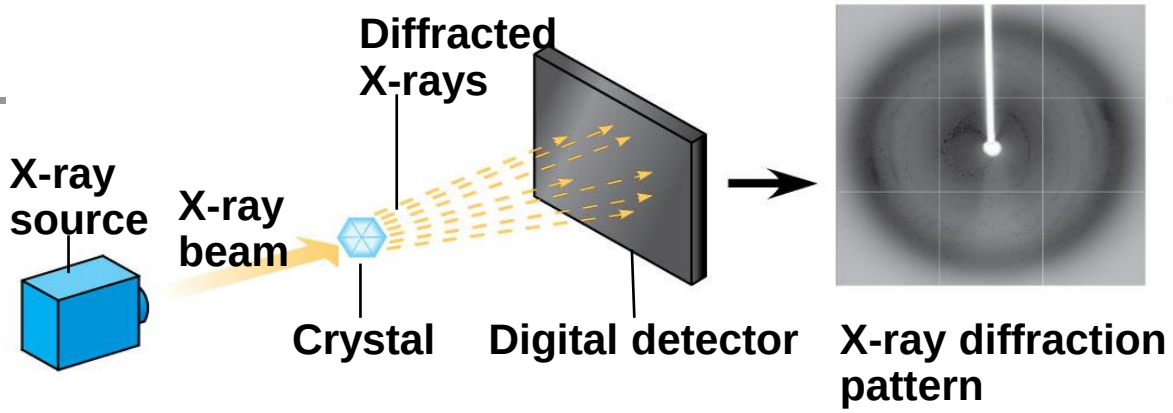
X-ray crystallography



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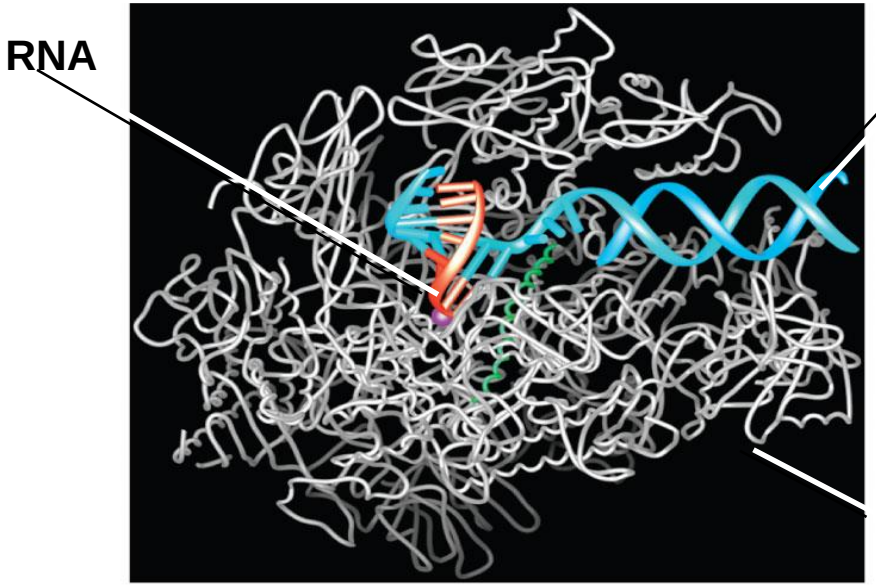

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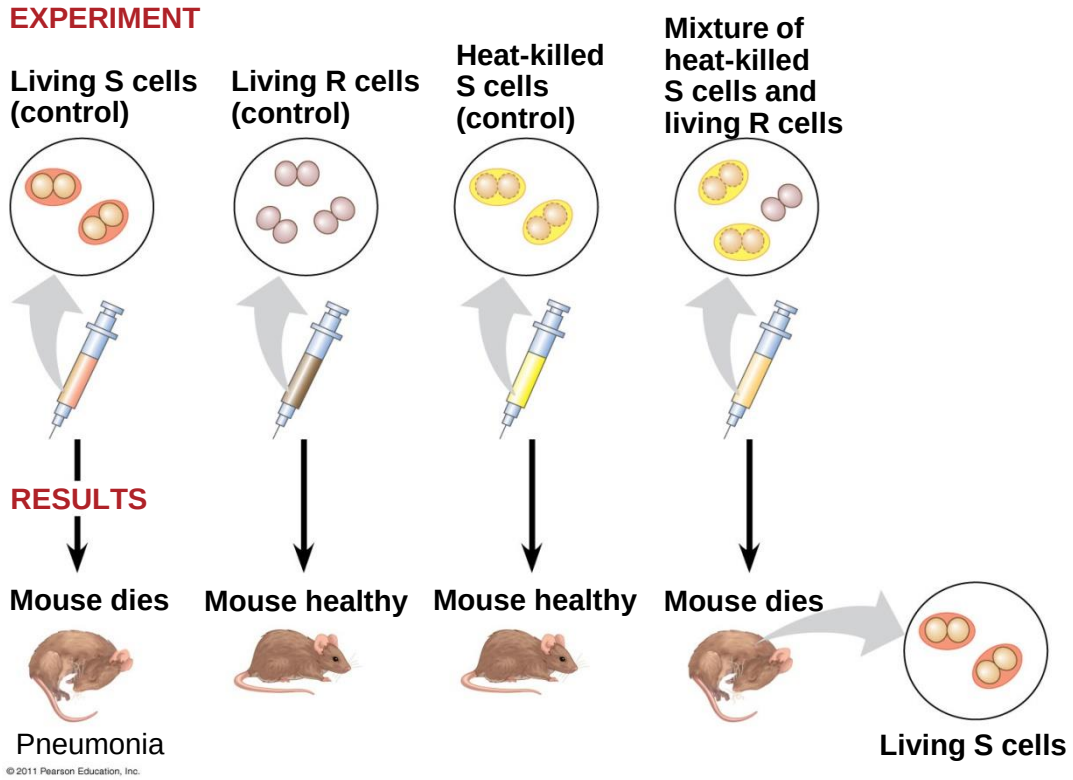
1928 **年**，格里菲斯用两种细菌菌株（一种生物物种的两个亚型）进行了研究，**其中一种具有致病性**，另一种则无害。当把致病菌株经 60 摄氏度加热致死后的残留物与无害菌株的活细胞混合时，
一些活细胞变成了致病细胞。他将这种现象称为转化（**从** S **型菌株到** R **型菌株的某种物**质转移）。他们的结论是基于实验证据得出的，**即只有** DNA 能够将无害细菌转化为致病细菌，但当时这一结论并未得到广泛认可。

遗传物质

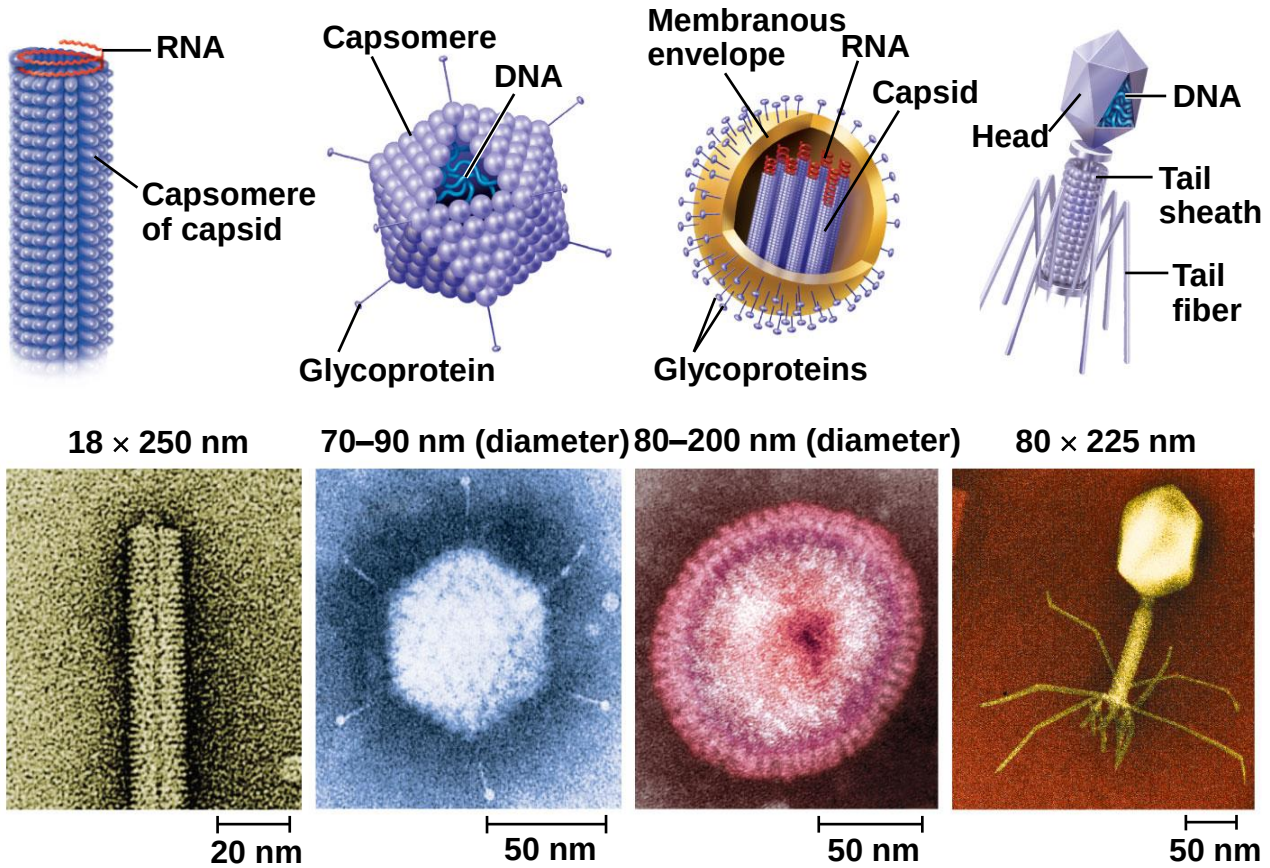
DNA is the genetic material

living s cell/killed s cell



- In 1928 Griffith worked with two strains of a bacterium (a subtype of a biological species), one pathogenic and one harmless
- When he mixed heat-killed (60°C) remains of the pathogenic strain with living cells of the harmless strain 菌株, some living cells became pathogenic. He called this phenomenon **transformation** 转化 (something is transferred from S to R).
- Their conclusion was based on experimental evidence that only DNA worked in transforming harmless bacteria into pathogenic bacteria and was not well accepted at that time.

病毒与感染
Virus and infection



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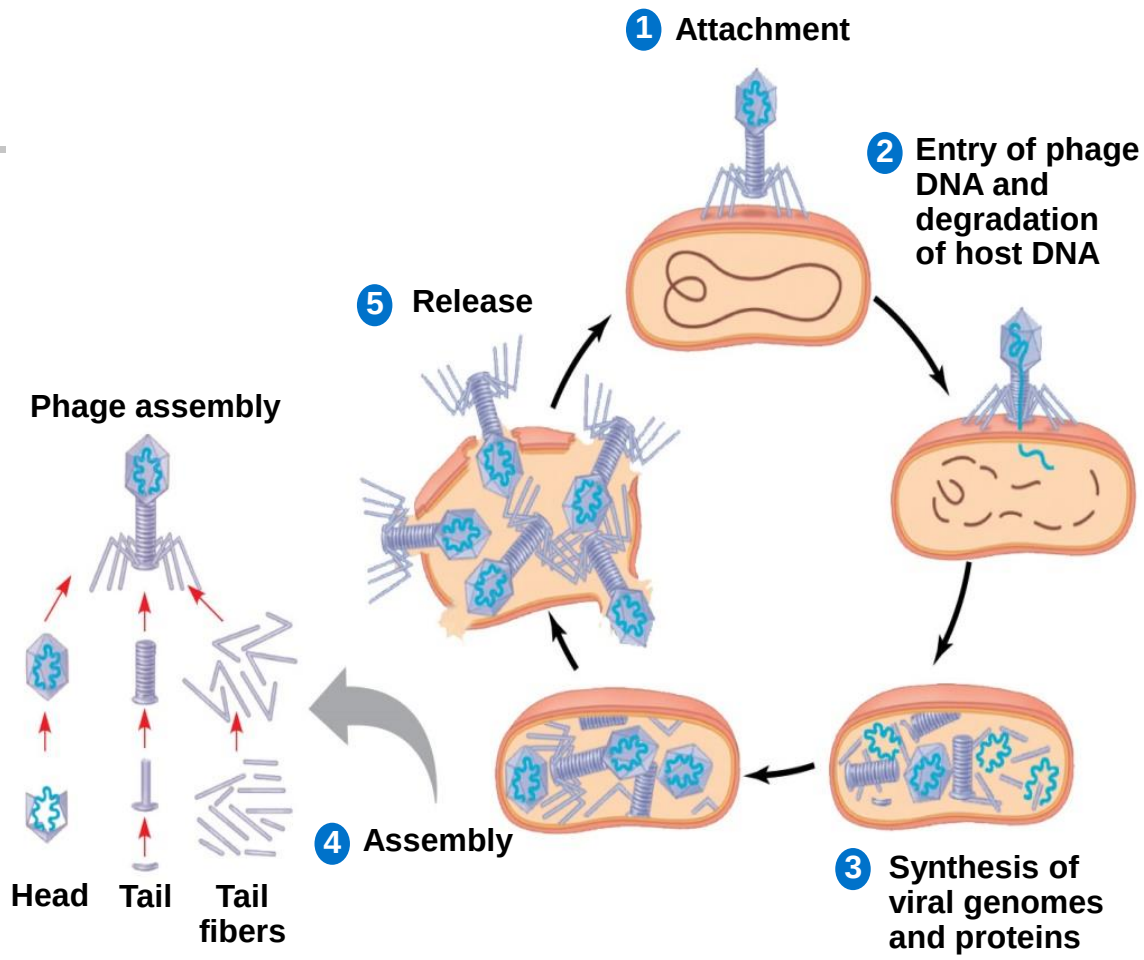

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Evidence That Viral DNA, not Protein, Can Program Cells

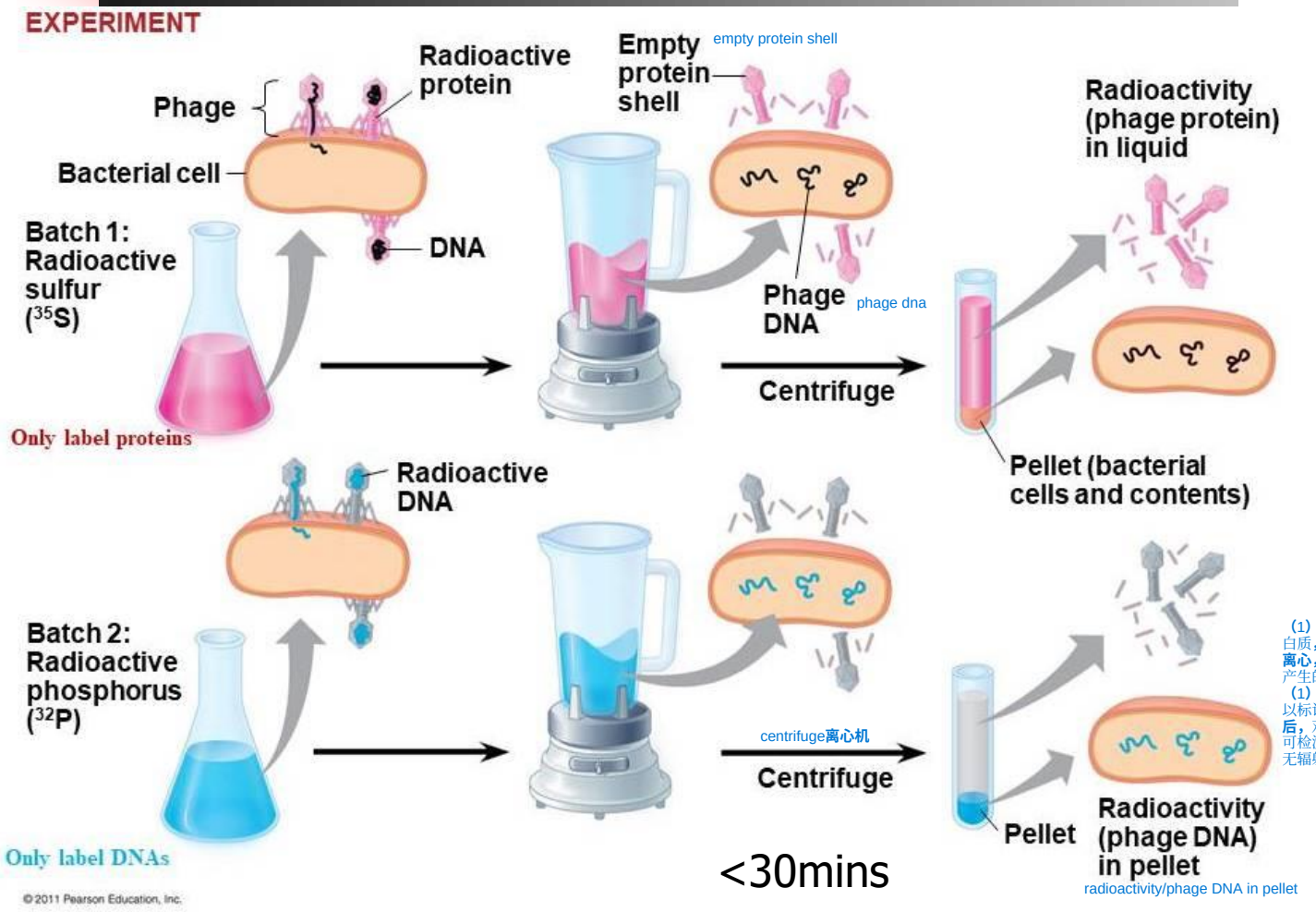
病毒 DNA 而非蛋白质可编程细胞的证据

噬菌体

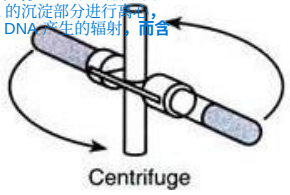
- Viruses, called **bacteriophages,** are widely used in molecular genetics research

(1) Grow the infected bacteria in ^{35}S solution to label proteins and purify the virus, (2) after ejecting of DNA into new bacteria and centrifugation, the soluble portion gives out detectable radiation produced by ^{36}S incorporating proteins of virus, while bacteria containing pellet does not.

(1) Grow the infected bacteria in ^{32}P solution to label nucleic acids of virus, (2) after ejecting of DNA into new bacteria and centrifugation of the pellet portion containing phage DNA gives out detectable radiation produced by ^{32}P labeled DNA ,while soluble port containing no radiation

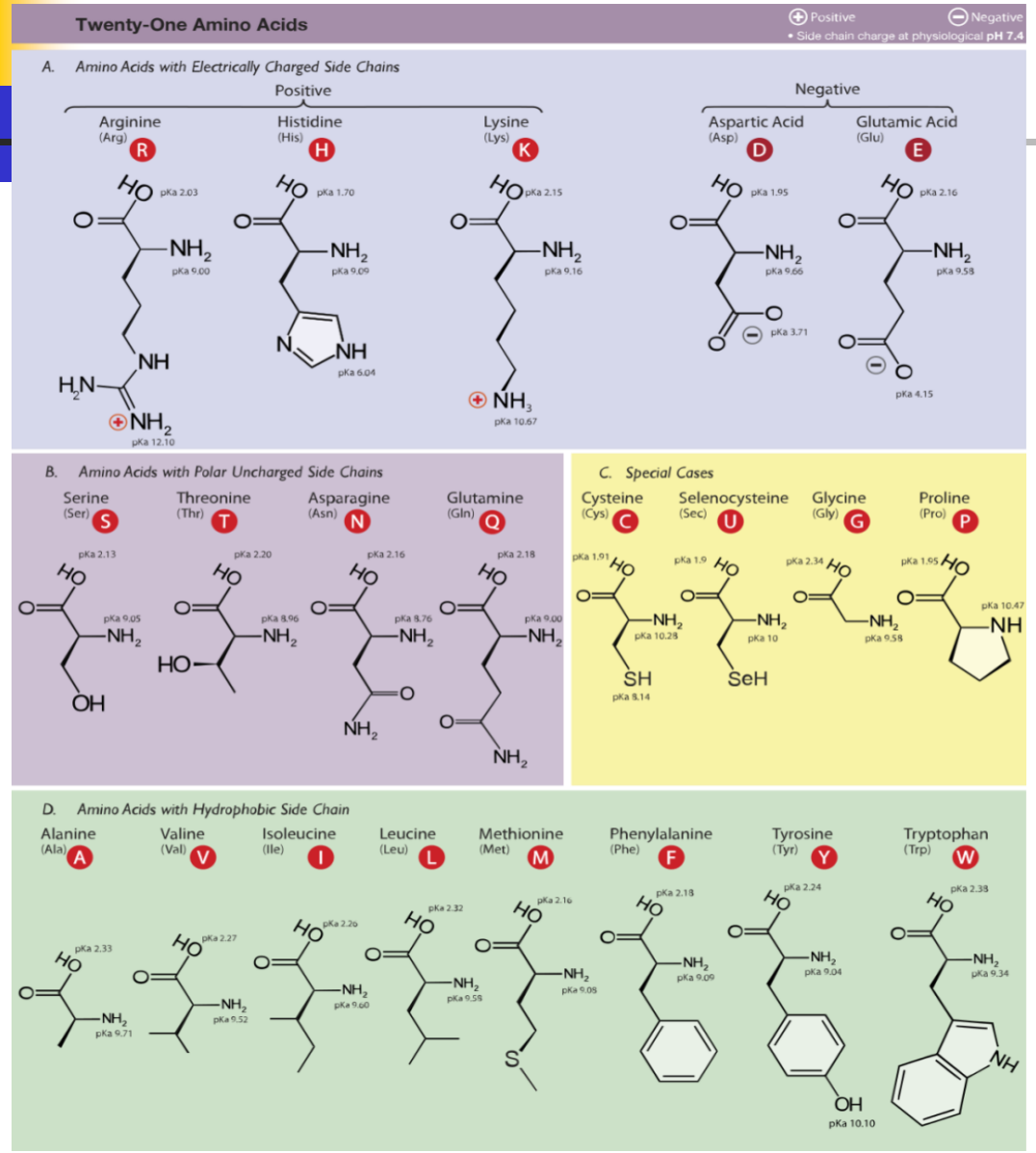


（1）将受感染的细菌在 35S 溶液中培养以标记蛋白质，并提纯病毒；（2）将 DNA 注入新细菌后离心，可溶部分会释放出由病毒蛋白质掺入 36S 所产生的可检测辐射，而含有沉淀物的细菌则不会。
（1）将受感染的细菌在含 32P 的溶液中培养，以标记病毒的核酸；（2）将 DNA 注入新细菌后，对含有噬菌体 DNA 的沉淀部分进行离心，可检测到由 32P 标记的 DNA 产生的辐射，而含无辐射的可溶部分。

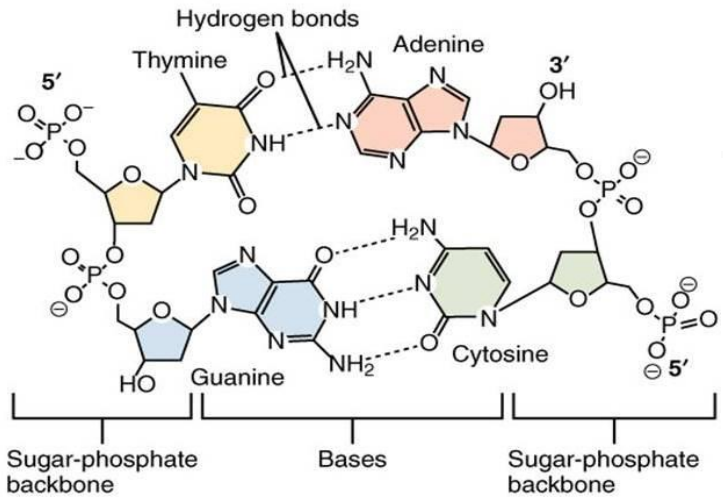


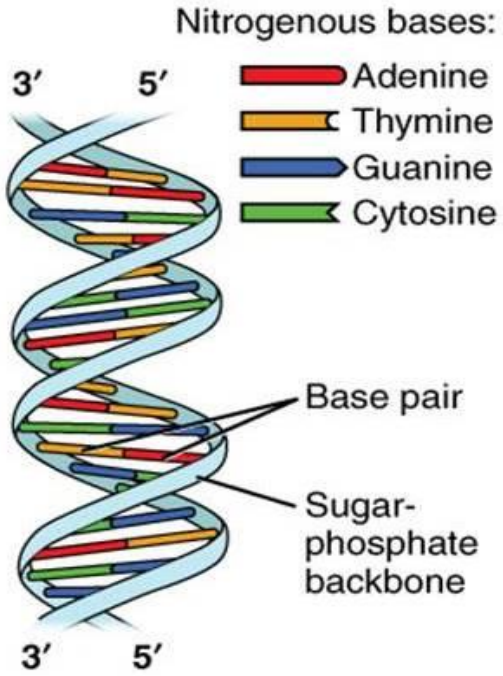
^{32}P and ^{36}S in DNA and Protein

Amino acids



nucleotides





构建 DNA 和 RNA 的结构模型

Building a Structural Model of DNA and RNA

Rosalind Franklin

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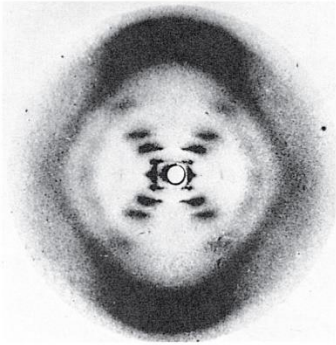
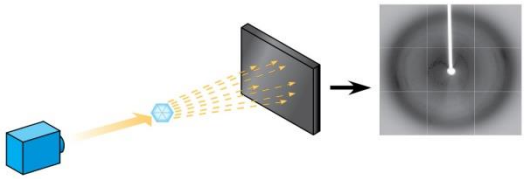
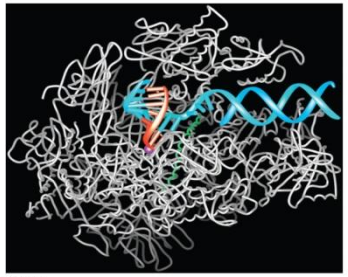


Franklin's X-ray diffraction photograph of DNA





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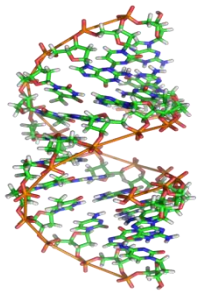
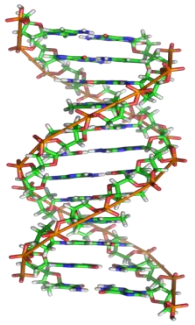
- Maurice Wilkins and Rosalind Franklin were using a technique called X-ray crystallography to study molecular structure
- Franklin produced a picture of the DNA molecule using this technique

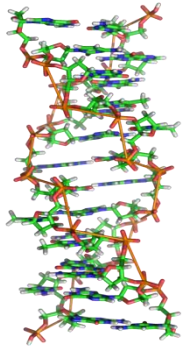
Structures of DNA and RNA



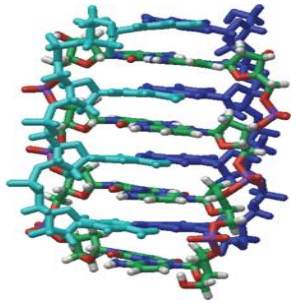
A-DNA
11bp
RH



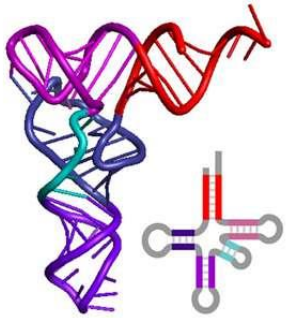
B-DNA
10bp
RH



Z-DNA
12bp
LF

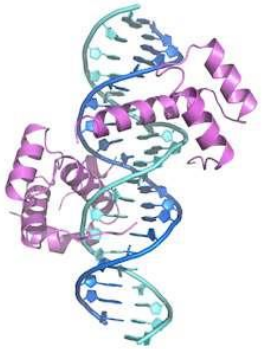


G-quadruplex DNA
4Gs



tRNA

mRNA

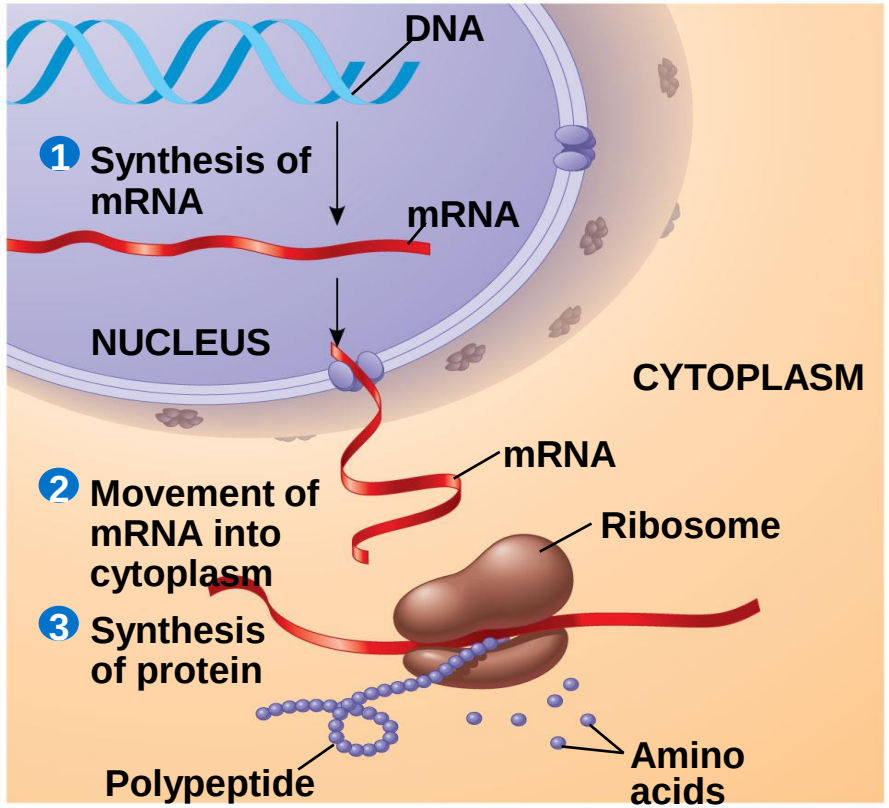


Protein/DNA complex

核酸储存、传递并帮助表达遗传信息。

Nucleic acids store, transmit, and help express hereditary information

多肽的氨基酸序列是由一种被称为基因的遗传单位所决定的（**基因是** DNA **的一段，**它编码具有功能的蛋白质）
基因由脱氧核糖核酸（DNA）**构成，而** DNA **是一种核苷酸聚合物，**其单体称为核苷酸。
核酸有两种类型:脱氧核糖核酸（DNA）**/核糖核酸**（RNA）DNA 为自身的复制提供指令
DNA 指导信使核糖核酸（mRNA）**的合成，**并通过 mRNA 控制蛋白质的合成
蛋白质的合成发生在核糖体上



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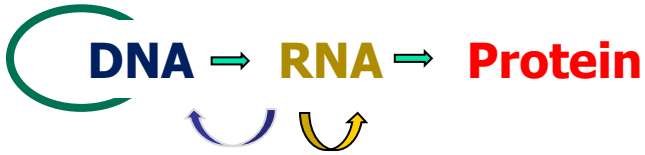
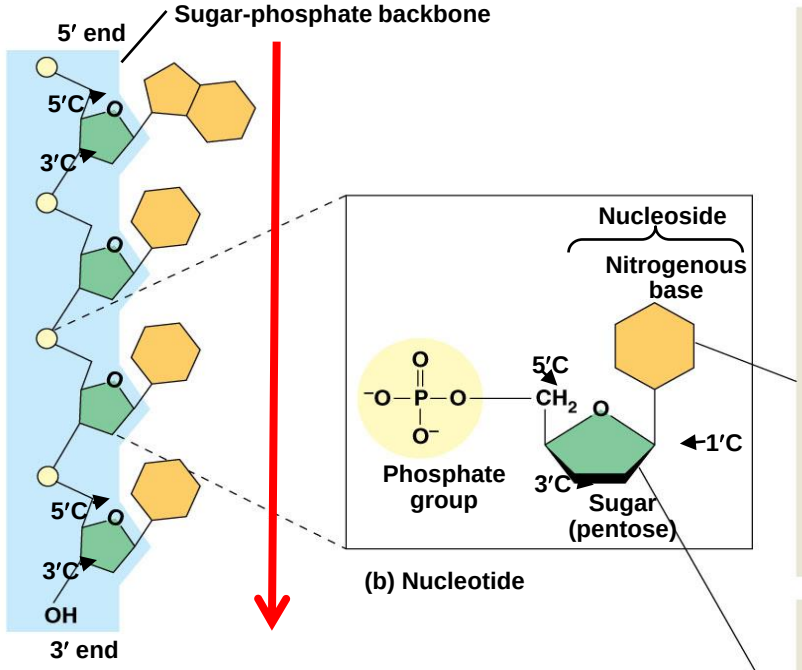
- The amino acid sequence of a polypeptide is programmed by a unit of inheritance called a **gene** (A gene is a segment of DNA that encodes function→ a protein)
- Genes are made of DNA, a **nucleic acid** made of monomers called nucleotides
- There are two types of nucleic acids
  - **Deoxyribonucleic acid (DNA)**
  - **Ribonucleic acid (RNA)**
- DNA provides directions for its own replication
- DNA directs synthesis of messenger RNA (mRNA) and, through mRNA, controls protein synthesis
- Protein synthesis occurs on ribosomes

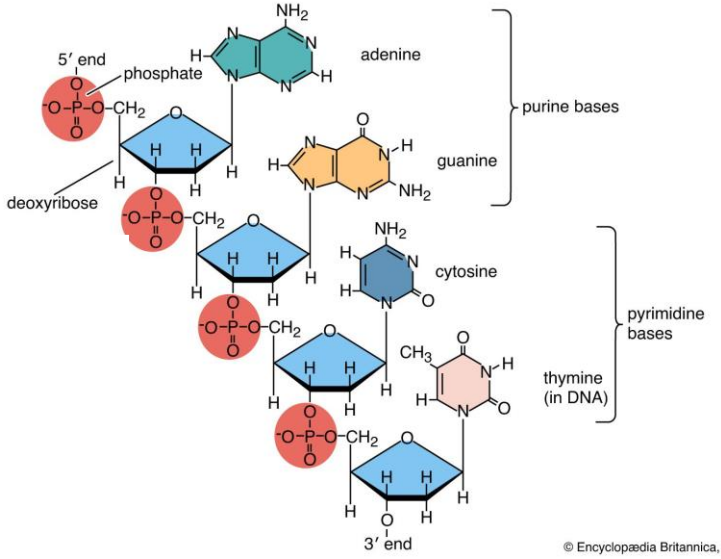


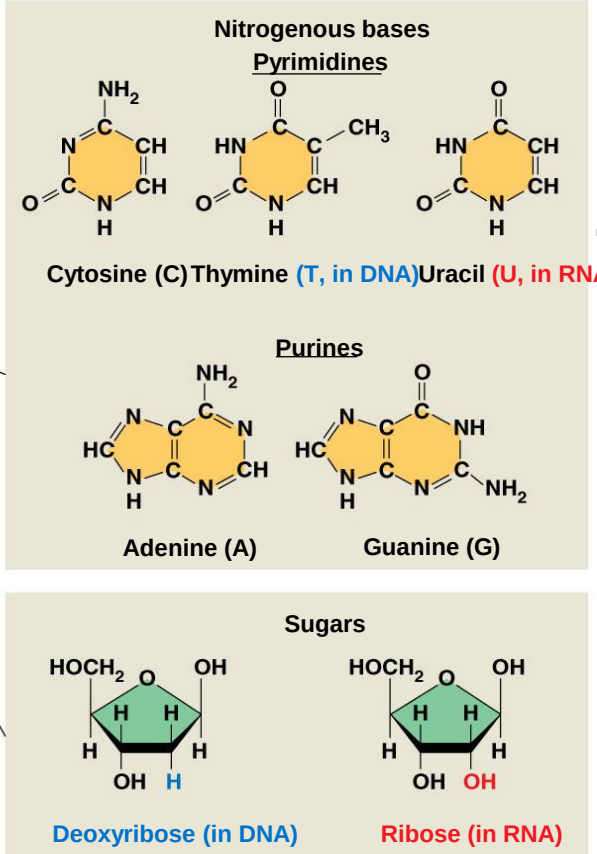
The Components of Nucleic Acids 核酸的组成成分



(a) Polynucleotide, or nucleic acid

(b) Nucleotide





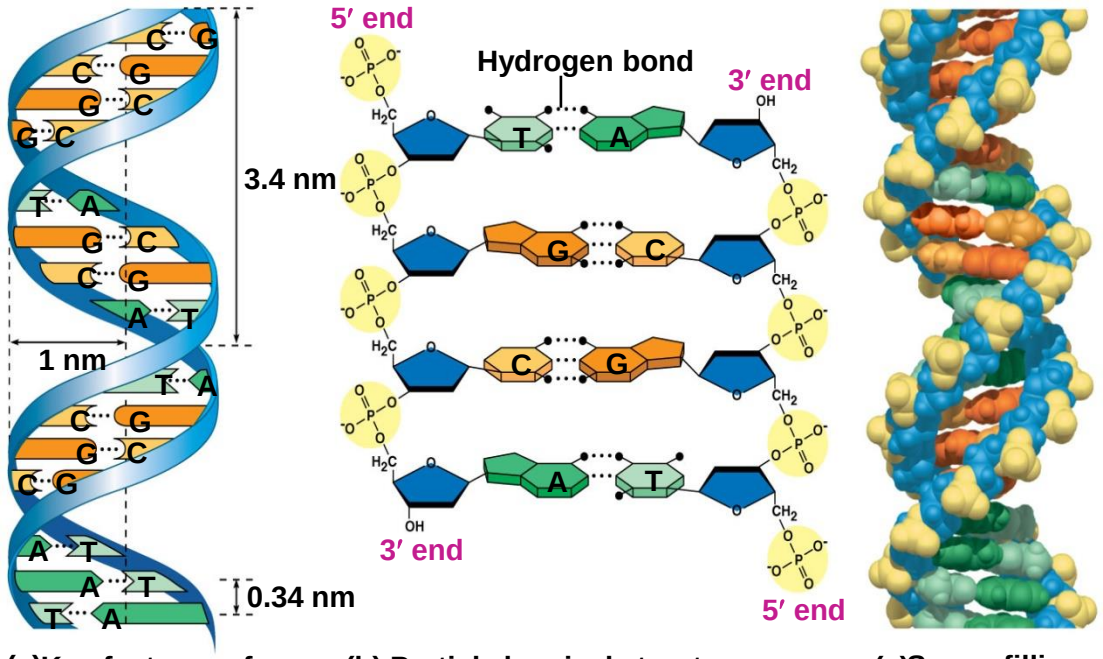
(c) Nucleoside components

- Nucleic acids are polymers called ***polynucleotides*** 多核苷酸
- Each polynucleotide is made of monomers called ***nucleotides*** 核苷酸
- Each nucleotide consists of a ***nitrogenous base***, ***a pentose sugar***, and one or more ***phosphate groups*** 含氮碱基 戊糖五糖
- The portion of a nucleotide without the phosphate group is called a ***nucleoside***
- Nucleoside = nitrogenous base + sugar
- There are two families of nitrogenous bases
  - 嘧啶 **Pyrimidines** (cytosine, thymine, and uracil) have a single six-membered ring
  - 嘌呤 **Purines** (adenine and guanine) have a six-membered ring fused to a five-membered ring
- In DNA, the sugar is **deoxyribose (ATGC)** ; in RNA, the sugar is **ribose (AUGC)**
- Nucleotide = nucleoside + phosphate group

Nucleotide Polymers (Facts)

- Nucleotides are linked together to build a polynucleotide
- Adjacent nucleotides are joined by covalent bonds that form between the —OH group on the 3′ carbon of one nucleotide and the phosphate on the 5′ carbon on the next
- These links create a backbone of sugar-phosphate units with nitrogenous bases as appendages
- The ***sequence of bases along*** a DNA or mRNA polymer is unique for each gene
- RNA molecules usually exist as single polynucleotide chains
- DNA molecules have two polynucleotides spiraling around an imaginary axis, forming a **double helix**
- In the DNA double helix, the two backbones run in opposite **5′→ 3′ directions (positive direction)** from each other, an arrangement referred to as **antiparallel**
- One DNA molecule includes many genes
- The nitrogenous bases in DNA pair up and form hydrogen bonds: adenine (A) always with thymine (T), and guanine (G) always with cytosine (C) Called **complementary** (**A-T, G-C**) base pairing
- Complementary pairing can also occur between two RNA molecules or between parts of the same molecule
- In RNA, thymine is replaced by uracil (U) so A and U pair (**A-U, G-C**)

Additional Features



(a)Key features of DNA structure (b) Partial chemical structure (c)Space-filling model

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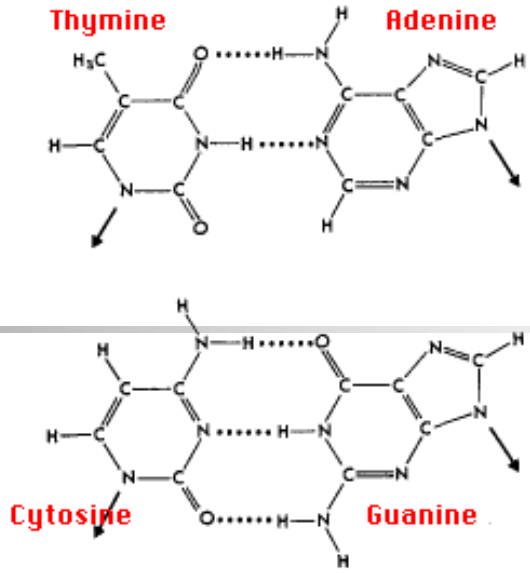
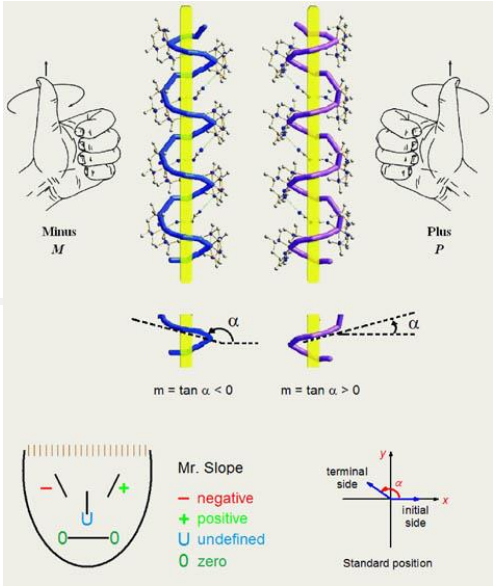
AT pair has two hydrogen bonds, while CG pair has three H-bond. More H-bonds → higher melting temperatures (more stable) (50-90 C)





Chargaff's rules

- The base composition of DNA varies between species
- In any species the number of A and T bases are equal, and the number of G and C bases are equal

The basis for these rules was not understood until the discovery of the double helix

5′–T A G G C C T–3′
3′–A T C C G G A–5′

Mismatch
5′–T A A G C C T–3′
3′–A T C C G G A–5′

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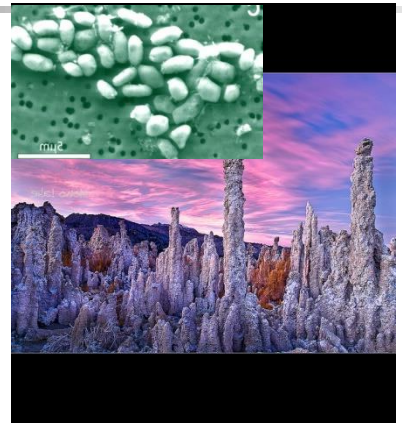
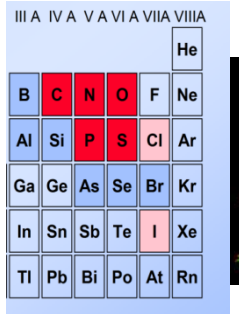
Interesting facts

How can shrimps survive in water 4 times hotter than boiling point?

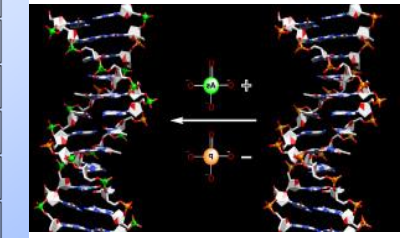
Thousands of the newly discovered species of shrimp. Shrimp mass around an oceanic hydrothermal vent beneath the Cayman Islands (5000m), in temperatures in excess of 450C (*Nat. Commu., 10 January 2012*)



Mono Lake, California



Arsenic-DNA



The newly discovered microbe, strain GFAJ-1, is a member of a common group of bacteria, the Gammaproteobacteria. In the laboratory, the researchers successfully grew microbes from the lake on a diet that was very lean on phosphorus but included generous helpings of arsenic. When researchers removed the phosphorus and replaced it with arsenic the microbes continued to grow. Subsequent analyses indicated that the arsenic was being used to produce the building blocks of new GFAJ-1 cells.(Science 2010)